Statement of interest:

- Recently, public debate over AI has intensified around job displacement, the energy demands of data centers, and existential risk. We are interested in this topic because we encounter these arguments constantly, in coursework, in news coverage, and in online discussion, and we have noticed that the same phrases recur across very different settings. That recurrence raises a question we cannot answer by reading alone: where does this vocabulary originate, and what happens to it as it travels?
- We also notice that public opinion on AI is forming faster than surveys can measure it. Surveys run a few times a year and capture attitudes after they settle, while the discourse producing those attitudes is continuous and public. We propose to analyze three corpora occupying different positions in the information chain, AI companies' blog posts, newspaper coverage, and Reddit discussion, to trace how framings emerge, travel, and change.

General problem/task definition: What are these papers trying to solve, and why?

- These papers try to understand public perception of AI through three different media outlets:
    - AI companies' blog posts: Anthropic, Google Deepmind, OpenAI
    - Newspaper: New York Times
    - Social media: Reddit (3 subreddits, r/technology, r/artificialintelligence, r/antiAI)
- We want to do this because public opinion of AI is currently being formed faster than it is measured. Survey instruments run only a few times a year, ask fixed questions, and capture attitudes after they are settled. The discourse around AI usually happens continuously and with the public, so it is scrapeable. Scraping these three sources allows us to observe real-time perception while it is formed.
- The three sources occupies a different position in the information chain:
    - AI companies' blog posts: upstream, initial framing, vocab, claims about their products. Everything downstream is a reaction to this.
    - Newspaper: professional, editorially constrained, aims at the public. Decides which companies' claims are news and they are presented.
    - Reddit: discussion, community-focused, pseudonymous. Good for capturing reasoning and disagreement, not just sentiment.

Main research question: How does AI discourse framing move from industry to press to public, and what changes at each step?

1. Propagation and lag: When a term, claim, or frame first appears in a lab blog post, how long until it appears in newspaper coverage, and then in Reddit discussion? Is that order consistent, or do some frames reach Reddit without passing through the press at all?
2. Narrative asymmetry: What important information did each source raise that the others ignore? For instance, capability announcements from companies versus labor displacement from Reddit.

Concise summaries of the articles:

1, Burns, Mary, et al. *A New Direction for Students in an AI World: Prosper, Prepare, Protect*. Brookings Institution, Jan. 2026, www.brookings.edu/wp-content/uploads/2026/01/A-New-Direction-for-Students-in-an-AI-World-FULL-REPORT.pdf.

- Summary: AI can be helpful in many different ways in educational settings, especially when it comes to bridging the resource gap, tailoring to students' needs, as well as optimizing teacher's time to achieve the greatest amount of result.
    - Bridging resource gap: For places where educational resources are underfunded, or having issues with absence of teachers, AI use in this setting can compensate for that by providing a tremendous amount of educational resources that are on par with the high

standard educational resources that exist elsewhere. Small language model or offline-capable architecture would also reduce the barrier of access, in terms of financial cost and hardware requirements, allowing children in underfunded areas to receive convenient, low-cost education

- Tailoring to students' needs: AI offers feedback and guidance on approaching a problem that is tailored specifically to individual students, depending on their differential in level of understanding and mastery of problems. In addition, AI if used properly can help create judgement-free environments for students as they navigate through challenges but at the same time concerned about teachers or peers' pressure
- Optimizing teachers' time: AI can help do routine tasks including course planning and structuring, automated scoring, translating materials, and writing draft emails for parents. Using AI this way can help save teachers tons of time over time, and this allows them to reinvest the time saved for more student engagement, communicating directly with students and their parents for certain things and providing helpful and necessary feedback.

These advantages of AI are built in the end upon using AI with guardrail; in the study, more participants are weighing AI to be causing more harm than benefit, 56% to 44%

- Undermining student's cognitive development: Developing brains require extensive effort-processing and mistake-making in order to build the neural pathways that help students develop as they grow up and encounter more difficult challenges. In addition, LLM's capacity to summarize stuff quickly will erode students' patience limit to remain focused when solving more complex problems, forcing them to lose patience and following instead on shallow reading
- Impeding social skills development: relying on AI chatbot, which usually has always–agreeable protocol in order to optimize users' satisfaction, will put students more likely to live in a rather simulated world and not focus on developing interpersonal skills that are more complex in actual world
- Eroding students' autonomy: Utilizing too much AI in this context actively facilitates students to not only develop the habit of shallow-reading but also pushing them to defer every task to AI for help. This erodes their capacity to develop judgment capacity, independent capabilities to develop studying skills, making students more and more reliant on AI instead of undergoing struggle that are beneficial for their brian development
- How to balance: AI for students should be deployed only when one is certain that this would help enhance students' learning rather than diminishing them, as this would involve not utilizing AI when it's time to push for students' cognitive development instead. In addition, in school settings, equip teachers with training to integrate AI into their teaching that can help manage class planning and capable of evaluating AI-generated content; teachers would also establish clear policy on AI that would ensure academic progress and equitable access

2, Clare, S., Prunkl, C., Bengio, Y., Bucknall, B. and Zhang, Y. (2026) International AI Safety Report 2026. Department for Science, Innovation and Technology (DSIT). Available at: https://internationalaisafetyreport.org/publication/international-ai-safety-report-2026 (Accessed: 11 October 2026).

- Summary: Since 2025, general purpose AI have improved their capabilities, with continued training that actively improves their performance. However at the same time, these AI tools can show a great differential in their effectiveness in handling different tasks. Occasionally, they can be seen to perform difficult tasks smoothly while in other times they fail at performing simpler tasks. With the growing development of AI, there is also growing concern about general AI use in public life. The concern mainly lies in areas including but not limited to public safety, malfunctioning, as well as risk to the labor market. For public safety, AI chatbot can be used in

harm's way, providing responses that are tailored to the needs of users, including but not limited to malicious intent such as generating content for scam email, blackmailing, and generating non-consensual imagery. In addition, other malicious intent used here would include using AI agent for scanning and searching for system vulnerabilities before they launch attacks on webpages; most harmfully, the same AI system can also be applied in settings like "democratizing" knowledge base for making bombs or chemical weapons that would pose direct threat to day-to-day life of the public. For malfunctions, oftentimes, one sees how AI, trying to be "agreeable" to its users, would help fabricating information and giving misguided advice. Similar to Brookings institute reports, if a person, like student in their developmental stage, develops reliance on the use of AI while avoiding to actually read and grow, fabrication of information or providing misguided advice in this context include risks where they could apply the same reliance when it comes to seeking professional healthcare support; misguided advice here can easily misguide people into consuming the wrong medication or receiving wrong treatment. At last, it's one widely discussed on how AI can pose a threat to the job market; in this case, with AI automating certain tasks, workers will lose more bargaining power when it comes to wage negotiations, leading to suppression of wages and shrinking employment sectors as a whole. Though slightly different from Brookings AI in education report that focus mainly on lower and secondary educational level, AI through streamlining certain tasks, can pose the risk to teachers' employment as they were then cut down with shorter working hours and thus leading to overall lower payment. At last, for this AI use, I want to see how the upper education setting is adapting to this AI development. Given so much of university administration is made of administrative tasks that involve routine. With the rise of AI, Brookings and "International AI safety report" should definitely extend their work to see how AI can contribute to lowering wages for administrative employees and eroding their wage-asking bargaining power.

3, Fast, Ethan, and Eric Horvitz. "Long-Term Trends in the Public Perception of Artificial Intelligence." Proceedings of the AAAI Conference on Artificial Intelligence, vol. 31, no. 1, 2017, doi:10.1609/aaai.v31i1.10635.

- Summary: This paper measures how AI has been portrayed in the press over three decades by analyzing New York Times articles published between January 1986 and mid-2016. The authors build a set of indicators covering engagement (how much AI is discussed), general optimism versus pessimism, and a fixed taxonomy of eight hopes (impact on work, education, transportation, healthcare, decision making, entertainment, positive singularity, positive human-AI merging) and seven concerns (loss of control, job displacement, military applications, absence of ethics, lack of progress, negative singularity, negative merging). Rather than relying on automated sentiment analysis, which they argue does not capture optimism, they segment articles into paragraphs, filter to those mentioning AI or robots, and pay Mechanical Turk workers to annotate roughly 8,000 paragraphs for relevance, attitude, and the presence of each hope or concern; these labels then bootstrap a classifier. Their main findings are that AI coverage fell during the 1987 AI winter, bottomed out in 1995, and rose sharply starting in late 2009; that coverage has consistently run two to three times more optimistic than pessimistic; but that specific fears, especially loss of control, ethical concerns, and negative effects on work, have climbed in recent years, while worries about slow progress have declined. Keyword analysis using mutual information shows the topics associated with AI shifting across five-year windows from space weapons to chess to search engines to driverless cars. To test whether press trends track the wider public, they apply their loss-of-control classifier to Reddit posts from 2010 to 2015 and find a matching upward curve.

4, La Cava, Lucio, et al. “Machines in the Crowd? Measuring the Footprint of Machine-Generated Text on Reddit.” Online Social Networks and Media, vol. 55, 2026, article 100355, https://doi.org/10.1016/j.osnem.2026.100355.

- Summary: The article examines Reddit and how AI generated text compares to human written content. The researchers analyzed 51 subreddits between 2022 and 2024 and analyzed their posts and comments using an AI detection method. Their findings were that AI-generated content remains relatively uncommon overall but can make up to 9% of the comments in certain communities. They found that AI-generated text was more prevalent in information seeking and social support communities. These texts tended to be longer, more repetitive, and more supportive in tone than human written text. Interestingly these comments often received similar or higher engagement than human comments. However, the researchers also acknowledged that AI detection is imperfect and their findings may not represent all of Reddit.

5, Maslej, N., Fattorini, L., Brynjolfsson, E., Etchemendy, J., Ligett, K., Lyons, T., Manyika, J., Ngo, H., Niebles, J.C., Parli, V., Shoham, Y., Wald, R., Clark, J. and Perrault, R. (2025) Artificial Intelligence Index Report 2025, arXiv:2504.07139 [Preprint]. doi:10.48550/arXiv.2504.07139. https://arxiv.org/abs/2504.07139

- Summary: This report examines how artificial intelligence has developed and expanded across business, education, healthcare, and everyday life. One finding that the article points out is that AI models are becoming more powerful, affordable, and widely accessible. Businesses are also investing in AI and more organizations are adopting them to improve productivity. The United States is continuing to lead in the development of advanced AI models, with China catching up quickly. Despite these improvements, the report raises concerns about AI bias, misinformation, privacy, environmental impact, and the difficulty in regulating this rapidly advancing technology.

6, Raghupathi, W., Ren, J. and Kulkarni, T. (2026) ‘Exploring the topics and sentiments of AI-related public opinions: An advanced machine learning text analysis’, *Information*, 17(2), p. 134. doi:10.3390/info17020134. **https://doi.org/10.3390/info1702013**

- Summary: The study investigates public opinion on AI by analyzing 28,819 Reddit comments posted between March 2015 and May 2024, collected from subreddits such as r/technology, r/news, and r/MachineLearning. The questions asked include: what topics dominate AI discussion, how sentiment has changed over time, and what language patterns separate positive from negative comments. To answer them, the authors combine 28 analytical techniques. The core ones include VADER, a sentiment tool designed for social media text; LIWC, which measures emotional and psychological word categories and is tested with regression to see whether those categories actually predict sentiment; and two topic-modeling methods, LDA and NMF, used to check each other’s results. These are supported other analysis such as tone. The average sentiment score generally fell, including after ChatGPT’s release in November 2022, with negative comments rising from 31.9% to 35.1%. The authors interprets that direct exposure to powerful AI intensifies public concern. Emotional words were the strongest predictors of whether a comment was positive or negative, suggesting people judge emotionally rather than technically. However, this could be a good direction for future research on AI governance.

7, ScienceDirect / Water Research article (2026) Water Research, S0043135426005488 [Preprint]. doi:10.1016/j.watres.2026.005488. https://www.sciencedirect.com/science/article/pii/S0043135426005488

- Summary: The paper offers a new lens for assessing AI: its water footprint, which has been overshadowed by electricity use and greenhouse gas emissions. Some may assume AI infrastructure such as data centers uses freshwater only for evaporative cooling, but it also uses water indirectly through electricity generation and the semiconductor manufacturing. Around two liters of water are needed per kWh of electricity, so even a modest 1 MW facility consumes about 25,500 cubic meters a year through evaporation. In the U.S., data center water usage increase, from 21.2 billion liters in 2014 to 66 billion in 2023. Projections suggest AI's global water footprint could reach up to 6.6 billion cubic meters annually by 2027. Location makes this worse: about two-thirds of data centers built or planned since 2022 are in water-stressed regions. For example, in Colon, Mexico, Microsoft built a hyperscale campus during a severe drought while nearby farming communities relied on trucked and bottled water. Some technologies already show large reductions. Meta's Sweden withdraws only about 25 million liters a year by using cold outside air. Their deployment remains limited by things like economic. The paper therefore introduces "digital water sobriety," a framework that evaluates water use for AI.

Synthesis (Compare and contrast):

- These sources fall into two groups. Fast and Horvitz, Raghupathi et al., and La Cava et al. study discourse – who is saying what about AI, and where. The other four are contributions to the discourse. The AI Index, the safety report, Brookings, and the water paper are the kind of upstream documents that newspapers cover and Reddit reacts to. That makes them useful twice over: as background for our literature review, and as examples of the upstream framing we want to trace. "Digital water sobriety" is a coined term with a known origin, and we could follow it downstream to see if it survives.
- Fast and Horvitz and Raghupathi et al. ask similar questions and answer them in opposite ways. Fast and Horvitz reject automated sentiment analysis and build a fixed taxonomy of hopes and concerns, then pay people to label paragraphs. Raghupathi et al. use VADER, LIWC, and topic models. The first approach is expensive but comparable across time; the second is cheap but produces categories that only mean something inside that corpus. This matters for us because our three corpora are written in very different registers. VADER was designed for social media and will probably misread a hedged, promotional blog post where the framing does the work, not the vocabulary. A consistent codebook applied to all three sources travels better.
- Their headline findings seem to conflict. Fast and Horvitz find press coverage two to three times more optimistic than pessimistic; Raghupathi et al. find sentiment falling and negative comments rising from 31.9% to 35.1%. But Fast and Horvitz stop in mid-2016, so the two may just be describing different periods. And their "optimism" is a human judgment about stance, while VADER measures word valence, which is not the same thing. We shouldn't treat the disagreement as a finding until we rule those out. Underneath, they actually agree: Fast and Horvitz find specific fears like loss of control and job displacement rising even while the overall ratio stays positive. They also already did a small version of what we propose, applying their loss-of-control classifier to Reddit from 2010 to 2015 and finding a matching curve. That's the closest precedent for our project, but it's one frame, coarse resolution, and before LLMs.
- The sources also show frames entering and leaving. Fast and Horvitz's fifteen categories include nothing environmental, but by 2025 the AI Index treats environmental cost as standard and the water paper builds a whole framework around it. Brookings worries that AI damages student cognition and autonomy, which falls under a category Fast and Horvitz coded as a hope. So a frame can keep its name and flip its valence, which is a reason not to borrow their codebook without changes
- Labor shows up almost everywhere but in different forms: the safety report treats it as workers losing bargaining power, Brookings as teachers' hours being cut, Raghupathi as an emotional

concern, since emotional word categories predicted sentiment better than topical ones. Same subject, but there exists three registers. Governance is where everyone converges: Reddit demands regulation, the AI Index notes it's hard, the water paper proposes a specific rule.

- The limitations stack up. Raghupathi and La Cava both note Reddit skews young, male, and technical, which applies directly to our subreddits. La Cava adds a worse problem: if up to 9% of comments in some communities are machine-generated, and that text is longer and gets similar or better engagement, then part of our corpus isn't public opinion. This limitation is unfortunately unsolved at the current moment. Since detection is imperfect, we can't just filter it out. And using only the NYT means we're measuring the Times, not the press.

Synthesis (Future work):

- The clearest gap is that nobody here studies industry communication as a corpus. Every paper treats company framing as an invisible background. Collecting lab blog posts is our main contribution, and it's what turns this from a sentiment study into a study of how things travel.
- Fast and Horvitz stop in mid-2016, before most of what people currently argue about. Extending their analysis to 2016-2026 would close that gap and test whether the optimism ratio survived 2022. It would also let us test their taxonomy directly: apply the original fifteen categories and measure how much recent coverage falls outside all of them.
- Their Reddit comparison shows two curves moving together but doesn't estimate lag. We'd want to find the first appearance of a term in each corpus, measure time to adoption at weekly resolution, and test whether the blog -> press -> Reddit order actually holds. The AI Index gives us an external timeline of releases and milestones to anchor the clock.
- The most interesting result would be a frame that skips the press entirely, or starts on Reddit and moves up. Water use, AI-attributed layoffs, and complaints about slop are plausible candidates.
- For method, we'd write one codebook, have an LLM do the bulk coding, and validate against a human-coded holdout, reporting agreement per category. A good test would be seeing whether that setup reproduces Fast and Horvitz's published curves, since those are a known answer. If it can't, we shouldn't trust it on new data. Following La Cava, we'd also run detection over the Reddit corpus and report results with and without flagged comments rather than claiming to have cleaned it.
- Finally, one outlet and one country is a real limit. Several outlets with different editorial stances would separate what the press does from what the Times does, and non-US coverage would show whether the pattern we find is general or just an artifact of an American industry talking to an American press (no current solution yet).

Reference

Note: we know the main contents should be 2 - 6 pages, so we reserve page 7 - end for the sources

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